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CYCLISTIC BIKE-SHARE CASE STUDY

How does a bike-share navigate speedy success?

I. INTRODUCTION

Welcome to the Cyclistic bike-share analysis case study! In this case study, I worked for a fictional company, Cyclistic, along with some key team members. In this data analysis process, I followed Six Phases: *Ask*, *Prepare*, *Process*, *Analyze*, *Share*, and *Act*.

II. SCENARIO

Assuming myself as a junior data analyst working on the marketing analyst team at Cyclistic, a bike-share company in Chicago. The director of marketing believes the company's future success depends on maximizing the number of annual memberships. Therefore, my team wants to understand how casual riders and annual members use Cyclistic bikes differently. From these insights, my team will design a new marketing strategy to convert casual riders into annual members. But first, Cyclistic executives must approve my recommendations, so they must be backed up with compelling data insights and professional data visualizations.

III. CHARACTERS AND TEAMS

- **Cyclistic:** A bike-share program that features more than 5,800 bicycles and 600 docking stations. Cyclistic sets itself apart by also offering reclining bikes, hand tricycles, and cargo bikes, making bike-share more inclusive to people with disabilities and riders who can't use a standard two-wheeled bike. The majority of riders opt for traditional bikes; about 8% of riders use the assistive options. Cyclistic users are more likely to ride for leisure, but about 30% use the bikes to commute to work each day.
- **Lily Moreno:** The director of marketing. Moreno is responsible for the development of campaigns and initiatives to promote the bike-share program. These may include email, social media, and other channels.
- **Cyclistic Marketing Analytics Team:** A team of data analysts who are responsible for collecting, analyzing, and reporting data that helps guide Cyclistic marketing strategy.
- **Cyclistic Executive Team:** The notoriously detail-oriented executive team will decide whether to approve the recommended marketing program.

IV. ABOUT THE COMPANY

In 2016, Cyclistic launched a successful bike-share offering. Since then, the program has grown to a fleet of 5,824 bicycles that are geotracked and locked into a network of 692 stations across Chicago. The bikes can be unlocked from one station and returned to any other station in the system anytime.

Until now, Cyclistic's marketing strategy relied on building general awareness and appealing to broad consumer segments. One approach that helped make these things possible was the flexibility of its pricing plans: single-ride passes, full-day passes, and annual memberships. Customers who purchase single-ride or full-day passes are referred to as casual riders. Customers who purchase annual memberships are Cyclistic members.

Cyclistic's finance analysts have concluded that annual members are much more profitable than casual riders. Although the pricing flexibility helps Cyclistic attract more customers, Moreno believes that maximizing the number of annual members will be key to future growth. Rather than creating a marketing campaign that targets all-new customers, Moreno believes there is a solid opportunity to convert casual riders into members. She notes that casual riders are already aware of the Cyclistic program and have chosen Cyclistic for their mobility needs.

V. Phase 1: ASK

Business Problem: Cyclistic aims to increase the number of annual memberships for future business growth, as annual members are more profitable than casual riders. Thus, the company wants to identify opportunities to convert casual riders into annual members.

Business Task: Analyze how annual members and casual riders use Cyclistic bikes differently.

Business Questions:

1. How do annual members and casual riders use Cyclistic bikes differently? [Primary Business Question]
2. Why would casual riders buy Cyclistic annual memberships?
3. How can Cyclistic use digital media to influence casual riders to become members?

Stakeholders:

- Lily Moreno
- Cyclistic Marketing Analytics Team
- Cyclistic Executive Team

VI. Phase 2: PREPARE

Data Source: The Cyclistic's historical trip data was collected from [Dataset Download](#). This is public data to explore how different customer types are using Cyclistic bikes.

The analysis uses 6 months of historical trip data covering *November 2025* through *April 2026*.

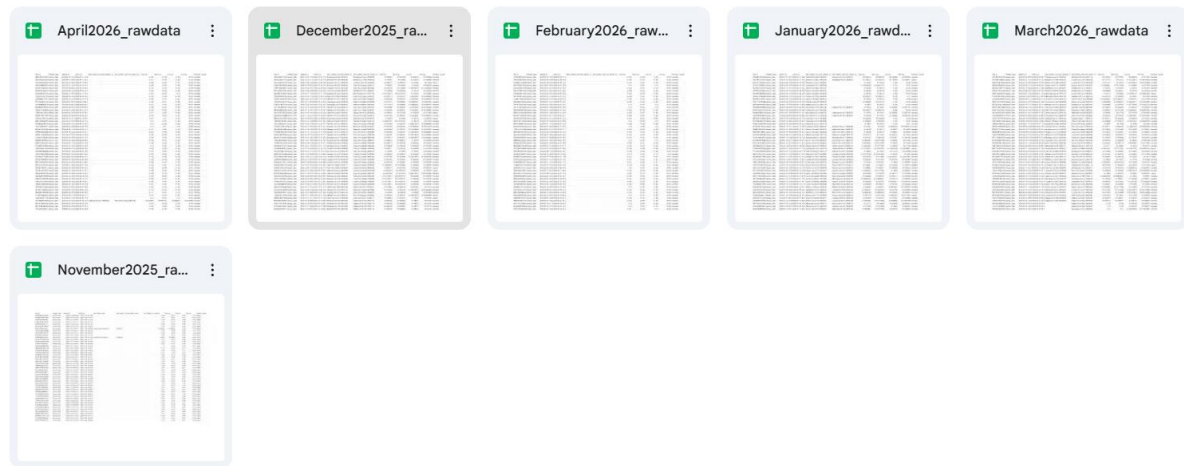


Fig. VI. 1. Monthly Cyclistic trip datasets successfully imported into Google Sheets for the period November 2025-April 2026.

Dataset Structure: The dataset includes 13 columns in each monthly dataset. The following table provides information about each column, its description, and its data type.

Table. VI. 1. Cyclistic Dataset Structure and Variables

Column	Description	Data Type
ride_id	Unique identifier assigned to each bike ride	Text
rideable_type	Type of bicycle used for the ride	Categorical
started_at	Date and time when the ride started	Date/Time
ended_at	Date and time when the ride ended	Date/Time

start_station_name	Name of the station where the ride started	Text
start_station_id	Identifier of the starting station	Text
end_station_name	Name of the station where the ride ended	Text
end_station_id	Identifier of the ending station	Text
start_lat	Latitude of the starting location	Numeric (Decimal)
start_lng	Longitude of the starting location	Numeric (Decimal)
end_lat	Latitude of the ending location	Numeric (Decimal)
end_lng	Longitude of the ending location	Numeric (Decimal)
member_casual	Indicates whether the rider is a member or casual rider	Categorical

Bias and Credibility: The data provides detailed trip-level information that is suitable for analyzing rider behavior and identifying trends over period of November 2025 to April 2026 . No specific sampling bias is identified in the available dataset information. Therefore, according to ROCCC framework the data is original from a reliable organization and it is comprehensive, current, and cited.

Data Ethics: The data has been made available by Motivate International Inc. under this [Data License Agreement](#) . This agreement includes all information about terms and conditions, Prohibited Conduct, Ownership and Provision of Data, and Termination of Agreement.

Data Integrity: The datasets were reviewed based on four key principles of data integrity: *Validity*, *Completeness*, *Accuracy*, and *Consistency*. These principles help ensure that the data is suitable for subsequent cleaning, transformation, and analysis.

VII. Phase 3: PROCESS

Google Spreadsheets was used as the initial tool to prepare the datasets for processing and analysis.

A. Data Cleaning *[Data cleaning was performed on copies of the raw datasets to preserve the original data files and ensure that the source data remained unchanged.]*

1. Spotting Errors

Process taken to highlight all cells that contain errors in each monthly spreadsheet:

Step 1: Click the empty rectangle (above row number 1 and left of column A) to select all cells in the spreadsheet.

Step 2: From the main menu, click “Format” and select “Conditional Formatting” to open the conditional format rules pane on the right.

Step 3: While in the ‘Single Color’ tab, under format rules, use the drop-down to select “Custom formula is”, enter “=ISERROR”, select any color (I chose red) for the formatting style, and then click “Done”.

Step 4: To remove conditional formatting, click “format” and select “Conditional formatting” and then click the “Trash” icon for the format rule.

Observation and Final Outcome: As the same process was followed for each monthly spreadsheet, the final observation for every monthly spreadsheet is that there were no errors.

2. Duplicate Records

Process taken to review duplicate records in each monthly spreadsheet:

Step 1: In the menu, select “Data”, then “Data clean-up”, and then “Remove duplicates”.

Step 2: Check the box next to “Data has header row”.

Step 3: Check the box next to “Select all” to inspect the entire spreadsheet.

Step 4: Finally click-on “Remove duplicates”.

Observation and Final Outcome: As the same process was followed for each monthly spreadsheet, the final outcome for every spreadsheet returned the

message “No duplicate rows were found”, indicating that no duplicate records were identified across the monthly datasets.

3. Identifying Missing Values

Process taken to identify columns containing missing values in each monthly spreadsheet: [Repeat this process for every column [Column A to Column M] in monthly spreadsheets.]

Step 1: Select the first column by clicking on ‘Column A’.

Step 2: In the menu, select “Data”, and then select “Create a filter”.

Step 3: Click on filter symbol beside Column A, in the “Filter by condition” dropdown select “Is empty” and click on “Ok”.

Step 4: This process allows you to identify the records which have missing values in the filter applied column. Note the column if it has returned any missing values.

Step 5: To remove the filter, go to menu and select “Data”, then click on “Remove filter”.

Observation and Final Outcome: As the same process was followed for each monthly spreadsheet, the final observation for every monthly spreadsheet resulted that the columns with missing values are start_station_name, start_station_id, end_station_name, end_station_id, end_lat, and end_lng.

B. Data Transformation:

1. Separate data into columns

Process taken to separate the data into different cells across each monthly spreadsheet: [The data in the started_at and ended_at columns contained both date and time. Using “Split text to columns”, the date and time values were separated into different cells, and the new columns were named started_at_date, started_at_time, ended_at_date, and ended_at_time.]

Step 1: Select started_at by clicking on Column C and right-click on selected region. Then click on “Insert 1 column to the right”, this creates a new empty column on the right of started_at.

Step 2: Again, select started_at by clicking on Column C.

Step 3: In the menu, select “Data” then click on “Split text to columns”.

Step 4: The delimiter will be automatically detected.

Observation and Final Outcome: As the same process was followed for each monthly spreadsheet, the final outcome for every spreadsheet is that the date data is remained in the *started_at* and the time data was inserted into the newly created empty column. Now rename the *started_at* column into *started_at_date* and the empty header to *started_at_time*.

Repeat the process for *ended_at* in Column E. After separating the data, rename the *ended_at* column to *ended_at_date* and the empty header to *ended_at_time*.

2. Format Dates and Times consistently

Process taken to change date data from number format to date format and time data from number format to time format cells across each monthly spreadsheet:

Step 1: Select *started_at_date* by clicking on Column C.

Step 2: From the menu, select “Format”, then select “Number” and then click on “Date”.

Step 3: Select *started_at_time* by clicking on Column D.

Step 4: From the menu, select “Format”, then select “Number” and then click on “Time”.

Observation and Final Outcome: As the same process was followed for each monthly spreadsheet, the final outcome for every spreadsheet is that the dates in the *started_at_date* and *ended_at_date* columns were formatted as date, while the times in the *started_at_time* and *ended_at_time* columns were formatted as time.

Repeat the process for *ended_at_date* and *ended_at_time* to change data from number format to date format and time format respectively.

3. Calculating the Length of Each Ride and Changing the Format to Time

Process taken to calculate ride_length and changing number format into time format for ride_length across each monthly spreadsheet: [Add a new column ride_length in Column P to calculate the length of each ride by subtracting the started_at_time from the ended_at_time.]

Step 1: Click on cell P1 and name the header as *ride_length*.

Step 2: Go to cell P2, add the formula **=F2-D2**, where F2 is *ended_at_time* and D2 is *started_at_time*, and then click Enter. It returns the length of ride.

Step 3: Double-click on the fill handle (small blue circle) to autofill other cells in column with the same formula in that cell.

Step 4: Select *ride_length* by clicking on Column P.

Step 5: From the menu, select “Format”, then select “Number” and then click on “Time”.

4. Calculate the day of the week that each ride started

Process taken to calculate the day of the week that each ride started across each monthly spreadsheet: [Add a new column day_of_week in Column Q.]

Step 1: Click on cell Q1 and name the header as *day_of_week*.

Step 2: Go to cell Q2, add the formula **=WEEKDAY(C2,1)**, where C2 is *started_at_date*, and then click Enter. It returns the day of the week that ride started.

Step 3: Double-click on the fill handle (small blue circle) to autofill other cells in column with the same formula in that cell.

Note: 1 - Sunday, 2 - Monday,, 7 - Saturday.

VIII. Phase 4: ANALYSIS

A. Data Preparation and BigQuery Setup

Downloaded each monthly dataset in .csv format. In BigQuery Sandbox, created a project named *cyclistic-case-study-da-01* and created a dataset named *cyclistic*. Then, created separate tables and uploaded the .csv file for each month. The tables were named *november_2025*, *december_2025*, *january_2026*, *february_2026*, *march_2026*, and *april_2026*.

SQL Query – Combining Monthly Datasets

Combined the six monthly tables into a single table named *combined_data* using SQL UNION ALL in BigQuery as following:

```
CREATE TABLE `cyclistic-case-study-da-01.cyclistic.combined_data` AS
SELECT * FROM `cyclistic-case-study-da-01.cyclistic.november_2025` UNION ALL
SELECT * FROM `cyclistic-case-study-da-01.cyclistic.december_2025` UNION ALL
SELECT * FROM `cyclistic-case-study-da-01.cyclistic.january_2026` UNION ALL
SELECT * FROM `cyclistic-case-study-da-01.cyclistic.february_2026` UNION ALL
SELECT * FROM `cyclistic-case-study-da-01.cyclistic.march_2026` UNION ALL
SELECT * FROM `cyclistic-case-study-da-01.cyclistic.april_2026`;
```

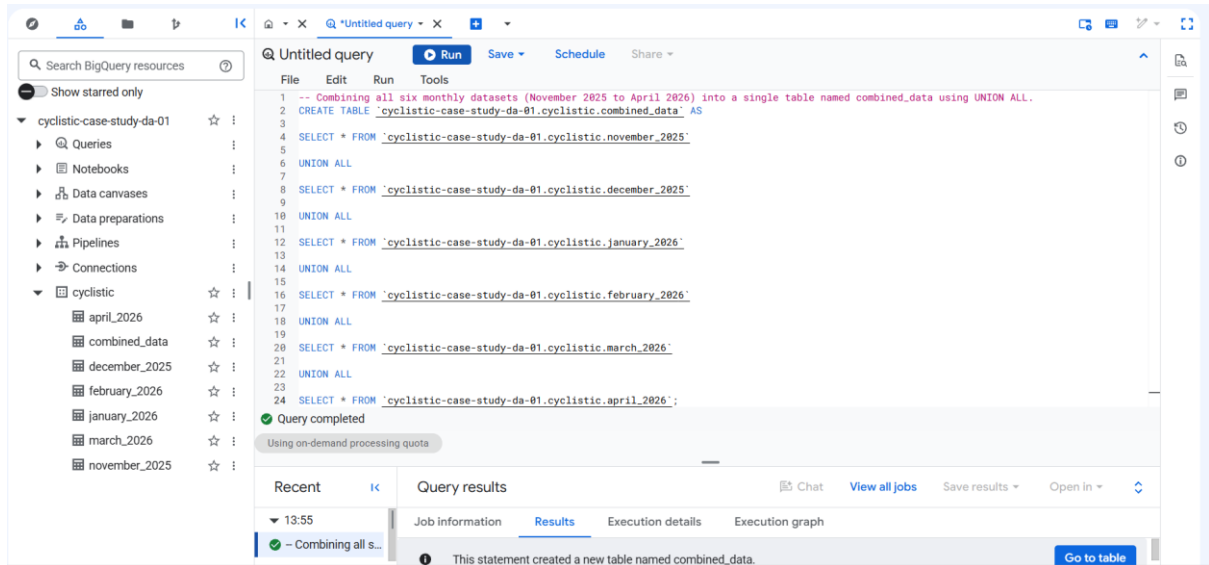


Fig. VIII. A. 1. SQL query used to combine the six monthly datasets into the combined_data table in BigQuery.

B. Exploring the Data

1. Checking the total number of rows

SQL Query:

```

SELECT COUNT(*) AS total_rides
FROM `cyclictic-case-study-da-01.cyclictic.combined_data`;

```

Output:

Recent	Query results	Chat	View all jobs	Save results	Open in
13:59	Job information	Results	Visualisation	JSON	Execution details
13:55	Row	total_rides			
	1	1601688			

Fig. VIII. B. 1. Snippet of the output showing the total number of records in the combined dataset.

Observation and Final Outcome: The combined dataset contains 1,601,688 ride records covering the period from November 2025 to April 2026.

2. Identifying distinct values

SQL Query:

```

SELECT DISTINCT *
FROM `cyclictic-case-study-da-01.cyclictic.combined_data`;

```

Output:

Row	ride_id	rideable_type	started_at_date	started_at_time	ended_at_date	ended_at_time
1	C9A0D69A2A069E5D	electric_bike	2026-02-24	19:38:03	2026-02-24	19:41:57
2	517CFE592EEE87E1	classic_bike	2025-12-20	15:41:20	2025-12-20	15:45:01
3	F1FF831D30F2CEC3	electric_bike	2026-01-30	09:22:59	2026-01-30	09:30:03
4	2E8E8E8893DE9B6A	electric_bike	2025-12-25	08:14:30	2025-12-25	08:20:26
5	5B25FFB127693881	classic_bike	2025-12-03	20:05:09	2025-12-03	20:06:59

Results per page: 50 1 - 50 of 1601688

Fig. VIII. B. 2. Snippet of the output showing the distinct records in the combined data.

Observation and Final Outcome: The combined dataset contains 1,601,688 DISTINCT ride records covering the period from November 2025 to April 2026.

Note: The total number of records is equal to the number of distinct records, indicating that there are no duplicate records in the combined dataset.

3. Calculating Mean of ride_length

SQL Query:

```
SELECT
ROUND(
  AVG(
    EXTRACT(HOUR FROM ride_length) * 60
    + EXTRACT(MINUTE FROM ride_length)
    + EXTRACT(SECOND FROM ride_length) / 60
  ), 2
) AS mean_ride_length_minutes
FROM `cyclistic-case-study-da-01.cyclistic.combined_data`;
```

Output:

Row	mean_ride_length...
1	12.29

Fig. VIII. B. 3. Snippet of the output showing the mean ride length in the combined dataset.

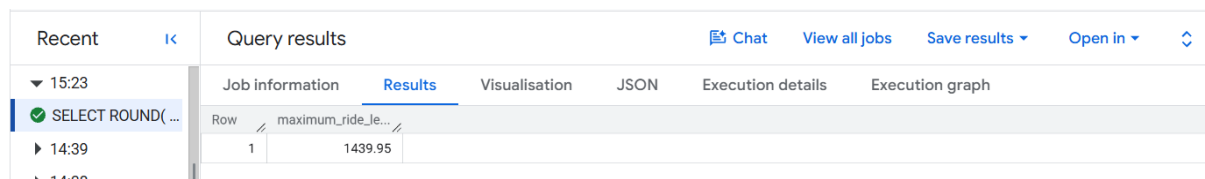
Observation and Final Outcome: The ride length was converted from time format into minutes for calculation, and the resulting mean ride length was rounded off to two decimal places. The mean ride length in the combined dataset is 12.29 minutes.

4. Calculating Max of ride_length

SQL Query:

```
SELECT
  ROUND(
    MAX(
      EXTRACT(HOUR FROM ride_length) * 60
      + EXTRACT(MINUTE FROM ride_length)
      + EXTRACT(SECOND FROM ride_length) / 60
    ), 2
  ) AS maximum_ride_length_minutes
FROM `cyclistic-case-study-da-01.cyclistic.combined_data`;
```

Output:

The screenshot shows a web-based SQL interface. On the left, a 'Recent' sidebar lists queries with timestamps. The main area, titled 'Query results', has tabs for 'Job information', 'Results' (selected), 'Visualisation', 'JSON', 'Execution details', and 'Execution graph'. The 'Results' tab displays a table with one row and one column, 'maximum_ride_length_minutes', containing the value '1439.95'.

Row	maximum_ride_length_minutes
1	1439.95

Fig. VIII. B. 4. Snippet of the output showing the max ride length in the combined dataset.

Observation and Final Outcome: The ride length was converted from time format into minutes for calculation, and the resulting max ride length was rounded off to two decimal places. The max ride length in the combined dataset is 1439.95 minutes.

5. Calculate the most frequently occurring day of the week using mode

SQL Query:

```
SELECT
  day_of_week,
  COUNT(*) AS ride_count
FROM `cyclistic-case-study-da-01.cyclistic.combined_data`
GROUP BY day_of_week
ORDER BY ride_count DESC
LIMIT 1;
```

Output:

Recent

Query results

Chat

View all jobs

Save results

Open in

▼ 16:18

Job informationResultsVisualisationJSONExecution detailsExecution graph

✓ – Identify the mo...

▶ 16:18

Row	day_of_week	ride_count
1	4	253717

Fig. VIII. B. 5. Snippet of the output showing the most frequently occurring day of the week in the combined dataset.

Observation and Final Outcome: The most frequently occurring day of the week is **Wednesday (day 4)**, with 253,717 rides. Therefore, Wednesday is the mode for the day_of_week variable in the combined dataset.

C. Analyzing Ride Patterns by Membership Type

1. Number of rides by membership type

SQL Query:

```
SELECT
  member_casual,
  COUNT(*) AS total_rides
FROM `cyclistic-case-study-da-01.cyclistic.combined_data`
GROUP BY member_casual
ORDER BY total_rides DESC;
```

Output:

Recent

Query results

Chat

View all jobs

Save results

Open in

▼ 16:52

✓ SELECT member...

▶ 16:18

▶ 16:18

Job information

Results

Visualisation

JSON

Execution details

Execution graph

Row

member_casual

total_rides

1

member

1189167

2

casual

412521

Fig. VIII. C. 1. Snippet of the output showing the number of rides by membership type.

Observation and Final Outcome: The combined dataset contains 1,189,167 rides by members and 412,521 rides by casual riders. Members account for the larger share of total rides during the analyzed period.

2. Average ride length by membership type

SQL Query:

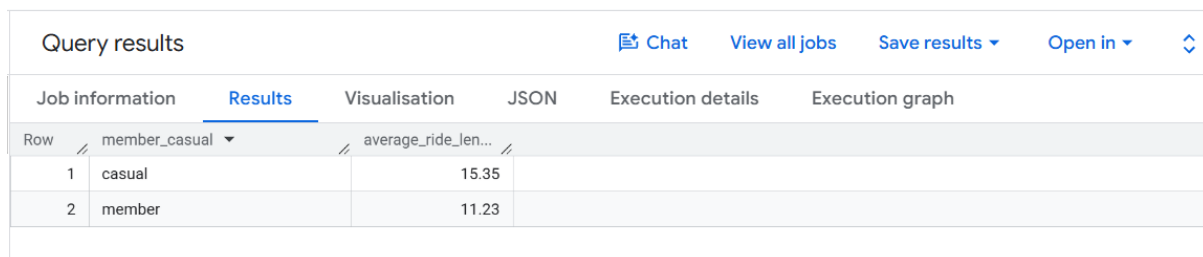
```
SELECT
```

```

member_casual,
ROUND(
  AVG(
    EXTRACT(HOUR FROM ride_length) * 60
    + EXTRACT(MINUTE FROM ride_length)
    + EXTRACT(SECOND FROM ride_length) / 60
  ), 2
) AS average_ride_length_minutes
FROM `cyclistic-case-study-da-01.cyclistic.combined_data`
WHERE ride_length IS NOT NULL
GROUP BY member_casual
ORDER BY average_ride_length_minutes DESC;

```

Output:



The screenshot shows a BigQuery interface with a query results table. The table has two columns: 'member_casual' and 'average_ride_length_minutes'. The first row is for 'casual' members with an average ride length of 15.35 minutes. The second row is for 'member' riders with an average ride length of 11.23 minutes. The interface includes tabs for 'Query results', 'Job information', 'Results', 'Visualisation', 'JSON', 'Execution details', and 'Execution graph'.

Row	member_casual	average_ride_length_minutes
1	casual	15.35
2	member	11.23

Fig. VIII. C. 2. Snippet of the output showing the Average ride length by membership type

Observation and Final Outcome: The average ride length was calculated in minutes and rounded off to two decimal places. Casual riders have an average ride length of 15.35 minutes, while members have an average ride length of 11.23 minutes. Therefore, casual riders have a higher average ride length than members.

D. Analyzing Ride Patterns by Day of Week

1. Number of rides by day of week

SQL Query:

```

SELECT
  day_of_week,
  COUNT(*) AS total_rides
FROM `cyclistic-case-study-da-01.cyclistic.combined_data`
GROUP BY day_of_week
ORDER BY total_rides DESC;

```

Output:

Query results				Chat	View all jobs	Save results ▼	Open in ▼	↕
Job information		Results	Visualisation	JSON	Execution details	Execution graph		
Row	day_of_week	total_rides						
1	4	253717						
2	5	252531						
3	3	249978						
4	6	232637						
5	2	222687						
6	7	216584						
7	1	173554						

Fig. VIII. D. 1. Snippet of the output showing the number of rides by day of the week.

Observation and Final Outcome: The number of rides was highest on **Wednesday (day 4)** with **253,717 rides**, followed by **Thursday (day 5)** with **252,531 rides**. The lowest number of rides was recorded on **Sunday (day 1)** with **173,554 rides**.

2. Average ride length by day of week

SQL Query:

```
SELECT day_of_week,
ROUND( AVG(
    EXTRACT(HOUR FROM ride_length) * 60
    + EXTRACT(MINUTE FROM ride_length)
    + EXTRACT(SECOND FROM ride_length) / 60
), 2 ) AS average_ride_length_minutes
FROM `cyclistic-case-study-da-01.cyclistic.combined_data`
GROUP BY day_of_week ORDER BY average_ride_length_minutes DESC;
```

Output:

Query results				Chat	View all jobs	Save results ▼	Open in ▼	↕
Job information		Results	Visualisation	JSON	Execution details	Execution graph		
Row	day_of_week	average_ride_len...						
1	1	14.18						
2	7	14.18						
3	6	12.37						
4	2	12.05						
5	5	11.43						
6	3	11.36						
7	4	11.31						

Fig. VIII. D. 2. Snippet of the output showing the average ride length by day of the week.

Observation and Final Outcome: The average ride length was calculated in **minutes** and rounded off to two decimal places. **Sunday (day 1)** and **Saturday (day 7)** recorded the highest average ride length at **14.18 minutes**, while **Wednesday (day 4)** recorded the lowest average ride length at **11.31 minutes**.

E. Analyzing Ride Patterns by Membership Type and Day of Week

1. Number of rides by membership type and day of week

SQL Query:

```
SELECT
  member_casual,
  day_of_week,
  COUNT(*) AS total_rides
FROM `cyclistic-case-study-da-01.cyclistic.combined_data`
GROUP BY member_casual, day_of_week
ORDER BY member_casual, day_of_week;
```

Output:

Query results					Chat	View all jobs	Save results ▾	Open in ▾	×
Job information		Results	Visualisation	JSON	Execution details		Execution graph		
Row	member_casual ▾	day_of_week ▾	total_rides ▾						
1	casual	1	60258						
2	casual	2	49333						
3	casual	3	50926						
4	casual	4	52439						
5	casual	5	57080						
6	casual	6	63746						
7	casual	7	78739						
8	member	1	113296						
9	member	2	173354						
10	member	3	199052						
11	member	4	201278						
12	member	5	195451						
13	member	6	168891						
14	member	7	137845						

Fig. VIII. E. 1. Snippet of the output showing the number of rides by membership type and day of the week.

Observation and Final Outcome: The number of rides was analyzed by membership type and day of the week. **Casual riders recorded the highest number of rides on Saturday (day 7) with 78,739 rides**, while their lowest number of rides was recorded on Monday (day 2) with 49,333 rides. **Members recorded the highest number of rides on Wednesday (day 4) with 201,278**

rides, while their lowest number of rides was recorded on Sunday (day 1) with 113,296 rides.

2. Average ride length by membership type and day of week

SQL Query:

```
SELECT member_casual,
       day_of_week,
       ROUND(
         AVG(
           EXTRACT(HOUR FROM ride_length) * 60
           + EXTRACT(MINUTE FROM ride_length)
           + EXTRACT(SECOND FROM ride_length) / 60
         ), 2
       ) AS average_ride_length_minutes
FROM `cyclistic-case-study-da-01.cyclistic.combined_data`
WHERE ride_length IS NOT NULL
GROUP BY member_casual, day_of_week
ORDER BY member_casual, day_of_week;
```

Output:

Query results Chat View all jobs Save results Open in

Job information	Results	Visualisation	JSON	Execution details	Execution graph
Row	member_casual	day_of_week	average_ride_len...		
1	casual	1	18.18		
2	casual	2	15.52		
3	casual	3	12.89		
4	casual	4	12.77		
5	casual	5	13.38		
6	casual	6	15.25		
7	casual	7	17.9		
8	member	1	12.06		
9	member	2	11.07		
10	member	3	10.97		
11	member	4	10.92		
12	member	5	10.86		
13	member	6	11.28		
14	member	7	12.06		

Fig. VIII. E. 2. Snippet of the output showing the average ride length by membership type and day of the week.

Observation and Final Outcome: The average ride length was calculated in minutes and rounded to two decimal places. *Casual riders recorded their highest average ride length on Sunday (day 1) at 18.18 minutes*, while their lowest was on Wednesday (day 4) at 12.77 minutes. *Members recorded their highest average ride length on Sunday (day 1) and Saturday (day 7), both at 12.06 minutes*, while their lowest was on Thursday (day 5) at 10.86 minutes. *Overall, casual riders had a higher average ride length than members across all days of the week.*

IX. Phase 5: SHARE

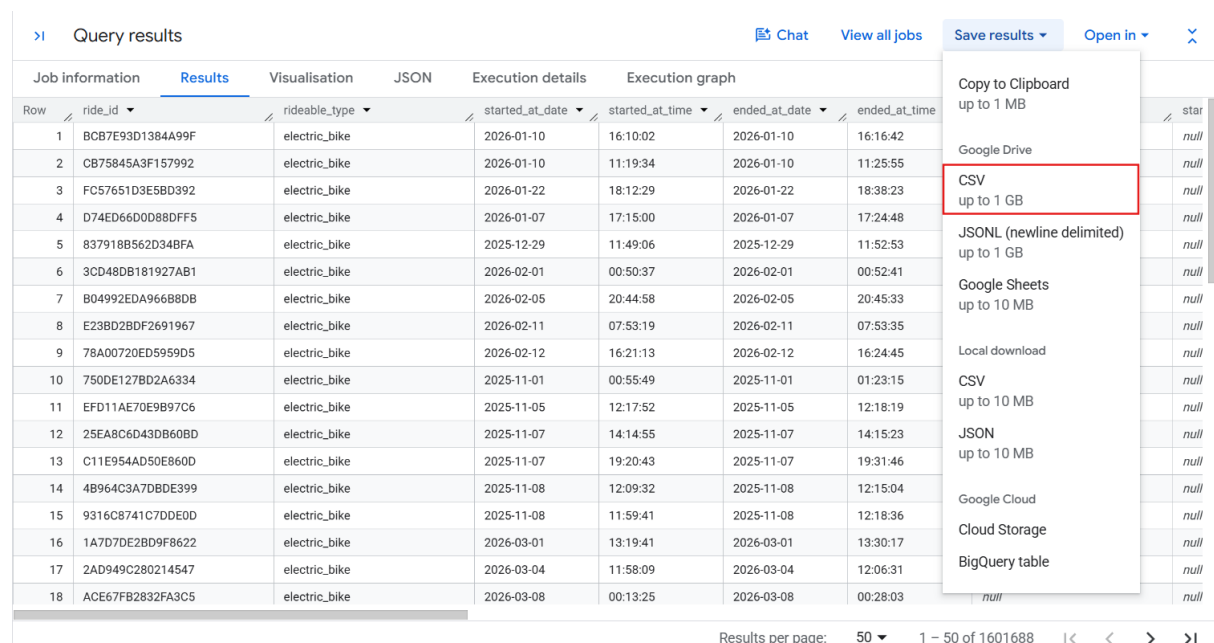
A. Preparing Data for Visualization

After completing the data analysis in BigQuery, the final combined dataset was downloaded in .csv format for visualization in Tableau Public. The combined_data table containing the six months of data from November 2025 to April 2026 was used as the source for the visualization.

The following SQL query was used to retrieve the complete combined dataset:

```
SELECT * FROM `cyclistic-case-study-da-01.cyclistic.combined_data`;
```

After executing the query, the query results were saved in .csv format and used as the data source for Tableau Public.



The screenshot displays the BigQuery 'Query results' interface. At the top, there are tabs for 'Job information', 'Results' (selected), 'Visualisation', 'JSON', 'Execution details', and 'Execution graph'. Below these tabs is a table with 18 rows of ride data. The columns are: Row, ride_id, rideable_type, started_at_date, started_at_time, ended_at_date, ended_at_time, and star. The 'Save results' dropdown menu is open, showing options: 'Copy to Clipboard up to 1 MB', 'Google Drive', 'CSV up to 1 GB' (highlighted with a red box), 'JSONL (newline delimited) up to 1 GB', 'Google Sheets up to 10 MB', 'Local download', 'CSV up to 10 MB', 'JSON up to 10 MB', 'Google Cloud', 'Cloud Storage', and 'BigQuery table'.

Row	ride_id	rideable_type	started_at_date	started_at_time	ended_at_date	ended_at_time	star
1	BCB7E93D1384A99F	electric_bike	2026-01-10	16:10:02	2026-01-10	16:16:42	null
2	CB75845A3F157992	electric_bike	2026-01-10	11:19:34	2026-01-10	11:25:55	null
3	FC57651D3E5BD392	electric_bike	2026-01-22	18:12:29	2026-01-22	18:38:23	null
4	D74ED66D0D88DFF5	electric_bike	2026-01-07	17:15:00	2026-01-07	17:24:48	null
5	837918B562D34BFA	electric_bike	2025-12-29	11:49:06	2025-12-29	11:52:53	null
6	3CD48DB181927AB1	electric_bike	2026-02-01	00:50:37	2026-02-01	00:52:41	null
7	B04992EDA966B8DB	electric_bike	2026-02-05	20:44:58	2026-02-05	20:45:33	null
8	E23BD2BDF2691967	electric_bike	2026-02-11	07:53:19	2026-02-11	07:53:35	null
9	78A00720ED5959D5	electric_bike	2026-02-12	16:21:13	2026-02-12	16:24:45	null
10	750DE127BD2A6334	electric_bike	2025-11-01	00:55:49	2025-11-01	01:23:15	null
11	EFD11AE70E9B97C6	electric_bike	2025-11-05	12:17:52	2025-11-05	12:18:19	null
12	25EA8C6D43DB60BD	electric_bike	2025-11-07	14:14:55	2025-11-07	14:15:23	null
13	C11E954AD50E860D	electric_bike	2025-11-07	19:20:43	2025-11-07	19:31:46	null
14	4B964C3A7DBDE399	electric_bike	2025-11-08	12:09:32	2025-11-08	12:15:04	null
15	9316CB741C7DDE0D	electric_bike	2025-11-08	11:59:41	2025-11-08	12:18:36	null
16	1A7D7DE2BD9F8622	electric_bike	2026-03-01	13:19:41	2026-03-01	13:30:17	null
17	2AD949C280214547	electric_bike	2026-03-04	11:58:09	2026-03-04	12:06:31	null
18	ACE67FB2832FA3C5	electric_bike	2026-03-08	00:13:25	2026-03-08	00:28:03	null

Fig. IX. A. 1. Query results of the combined_data table and the option to save the results in .csv format.

B. Uploading the Data to Tableau Public

The following steps were followed to upload the combined dataset to Tableau Public:

Step 1: Open the Tableau Public profile and click on My Profile.

Step 2: Select Create from the drop-down menu and click on Viz.

Step 3: In the new Tableau Public workbook, the Connect to Data dialog box appears.

Step 4: Go to File and select Upload from Computer.

Step 5: Select the downloaded combined data .csv file from the computer.

Step 6: Click Open/OK to upload the file.

Step 7: The combined dataset with 17 fields 16,01,688 rows is successfully uploaded and becomes available for creating visualizations.

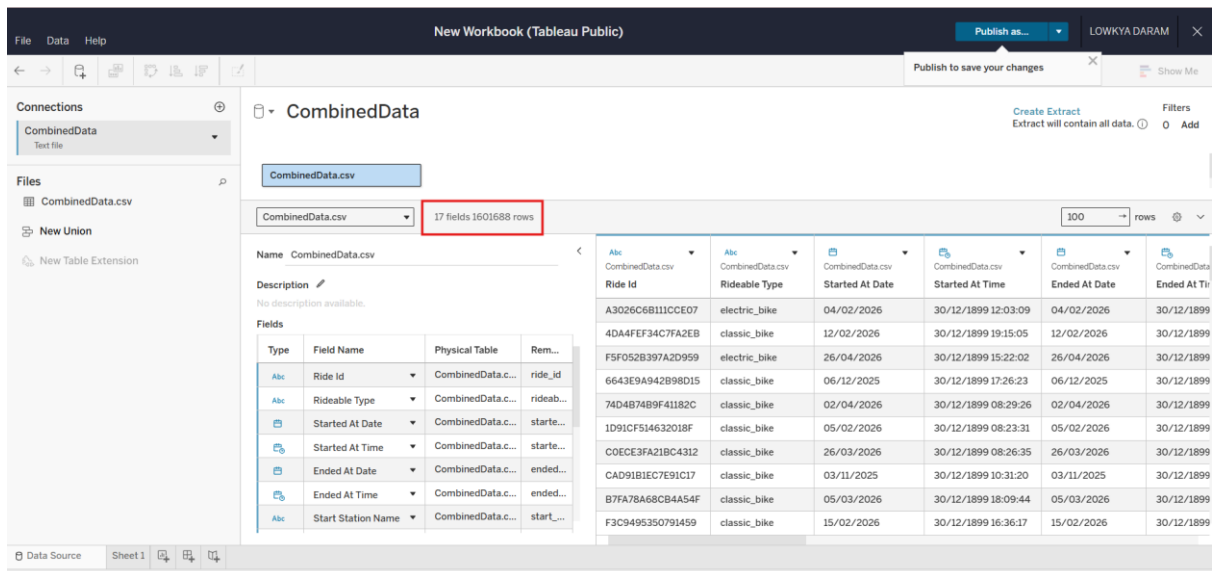


Fig. IX. B. 1. Combined dataset successfully uploaded to Tableau Public and connected to the workbook.

C. Data Visualization

Visualization 1: Number of Rides by Day of Week [Chart Type: Bar Chart]

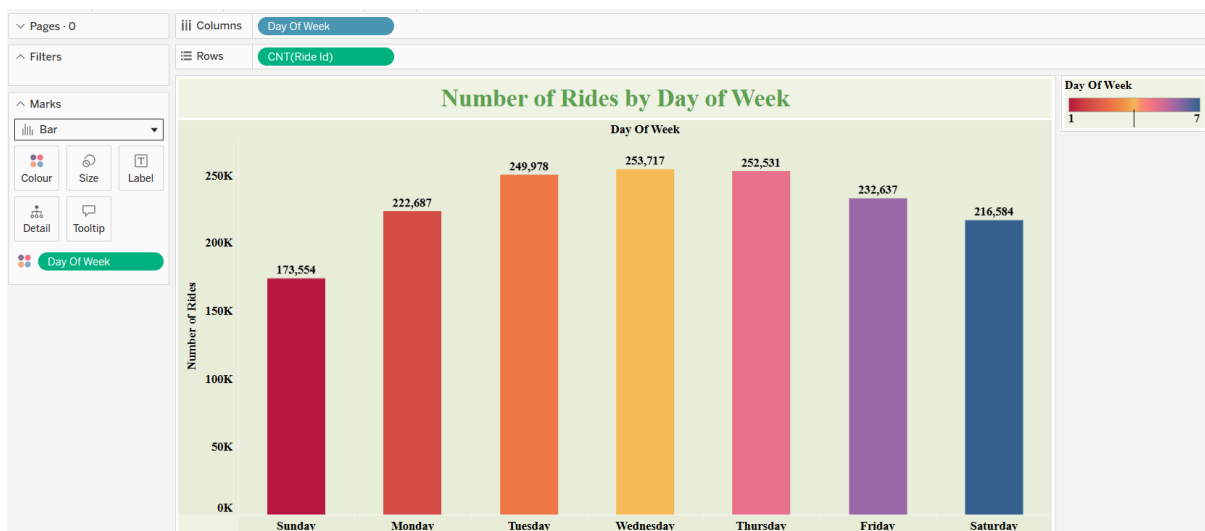


Fig. IX. C. 1. Number of rides by day of week from November 2025 to April 2026.

Description: The visualization shows the number of Cyclistic rides for each day of the week. Wednesday recorded the highest number of rides with 253,717 rides, while Sunday recorded the lowest with 173,554 rides.

Visualization 2: Number of Rides by Membership Type and Day of Week [Chart Type: Stacked Bar Chart]

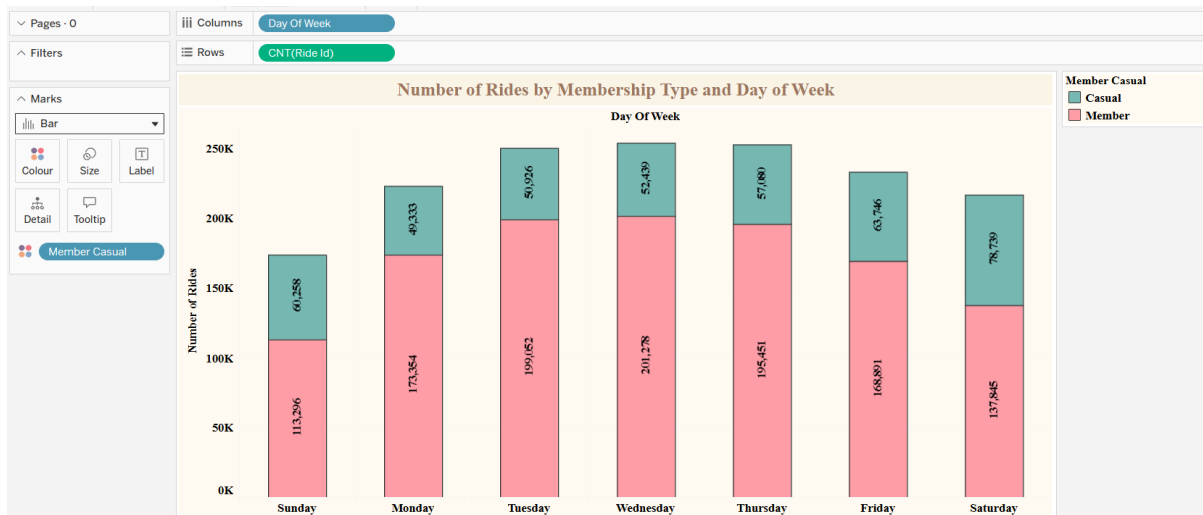


Fig. IX. C. 2. Number of rides by membership type and day of week from November 2025 to April 2026.

Description: The visualization shows the number of rides taken by members and casual riders across each day of the week. Members recorded more rides than casual riders on every day. Casual riders recorded their highest number of rides on Saturday (78,739), while members recorded their highest number on Wednesday (201,278).

Visualization 3: Monthly Number of Rides by Membership Type [Type: Line Chart]

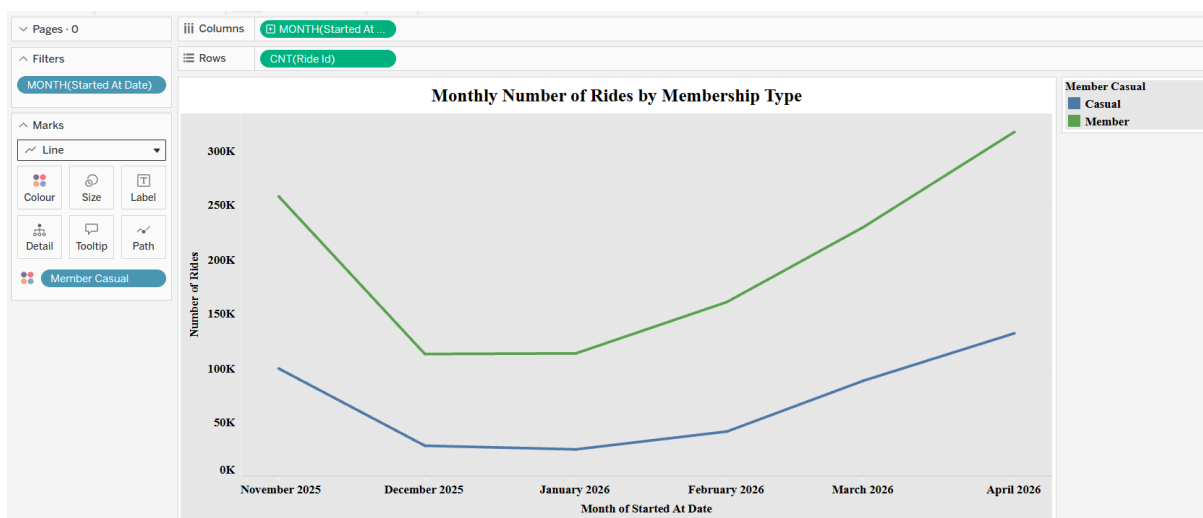


Fig. IX. C. 3. Monthly number of rides by membership type from November 2025 to April 2026.

Description: The line chart shows the monthly number of rides by membership type from November 2025 to April 2026. Casual riders showed comparatively lower monthly ride counts. The line chart shows that April has the highest number of rides, while January has the lowest number of rides.

Visualization 4: Average Ride Length by Membership Type and Day of Week [Type: Grouped Bar Chart]

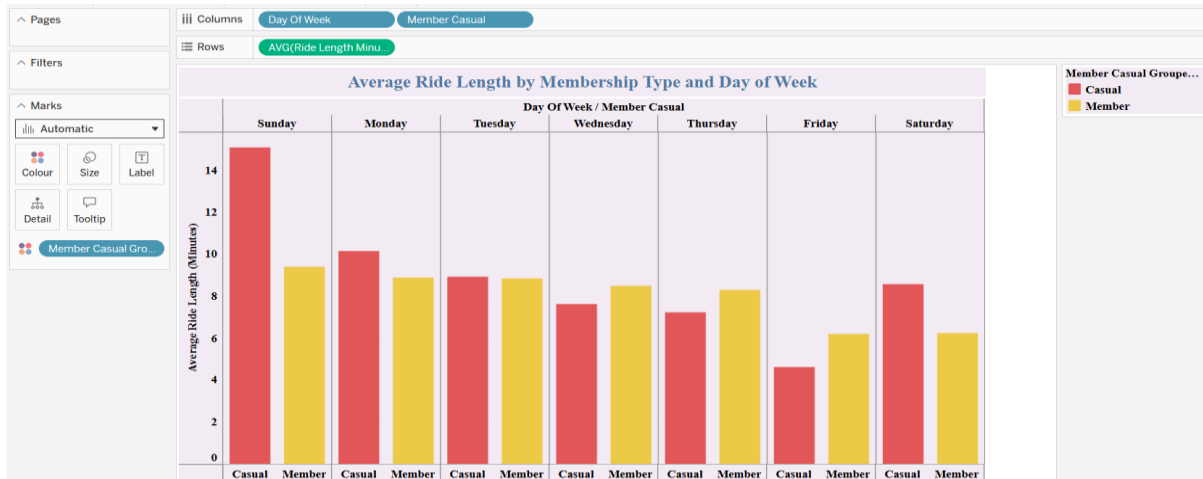


Fig. IX. C. 4. Average ride length by membership type and day of week from November 2025 to April 2026.

Description: The grouped bar chart compares the average ride length of casual riders and members across each day of the week. Casual riders had a higher average ride length on Sunday, Monday, Tuesday, and Saturday, with the highest average of 15.05 minutes on Sunday. Members had a higher average ride length on Wednesday, Thursday, and Friday, with the highest average of 8.47 minutes on Wednesday. Overall, casual riders recorded longer average rides on four days, while members recorded longer average rides on three days.

Visualization 5: Member Pie Chart [Type: Pie Chart]

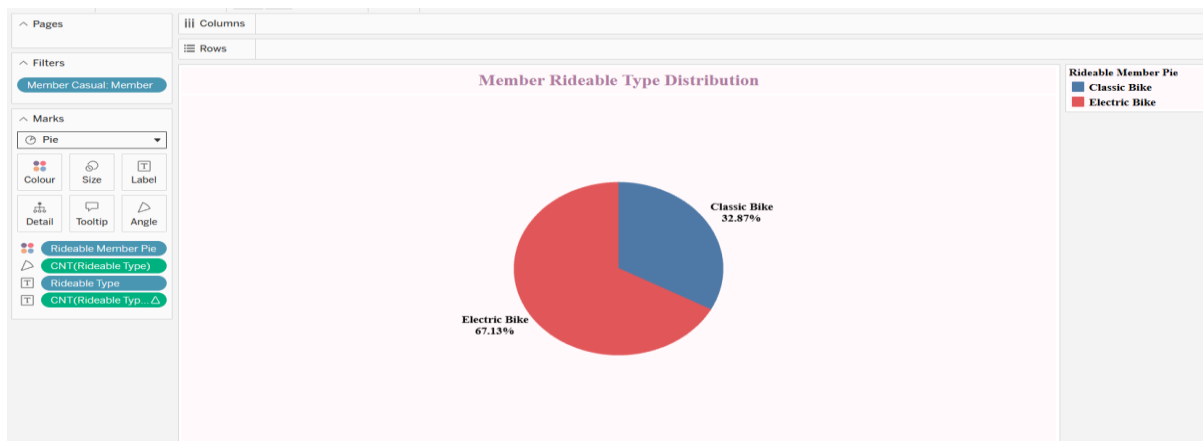


Fig. IX. C. 5. Member Rideable Type Distribution

Description: This pie chart illustrates how members are distributed across the different rideable types. The chart was created by filtering the dataset for Member users and using rideable type to represent the distribution of rides. The pie chart shows that electric bikes account for the highest percentage of rides among members, followed by classic bikes. This indicates that members have a stronger preference for electric bikes compared with classic bikes.

Visualization 6: Casual Pie Chart [Type: Pie Chart]

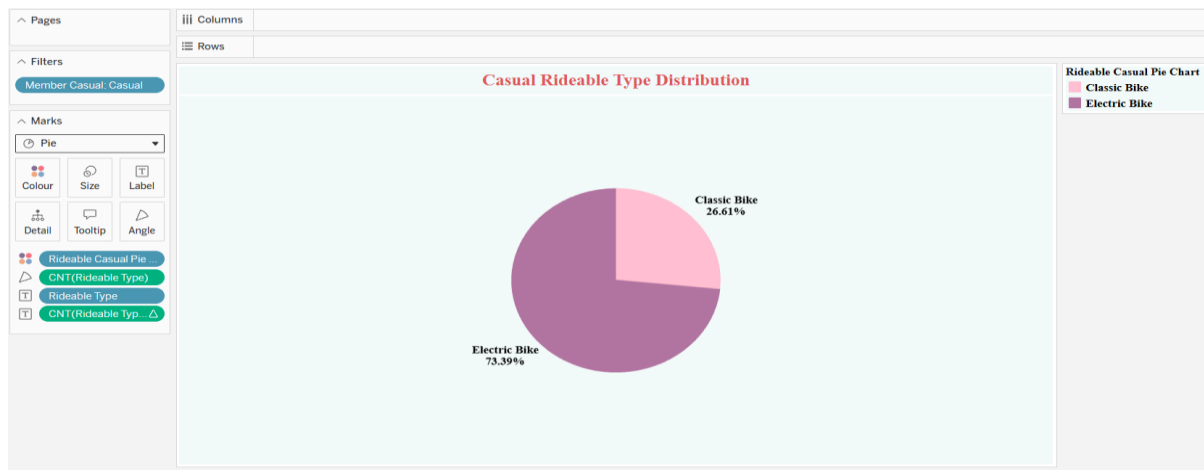


Fig. IX. C. 6. Casual Rideable Type Distribution

Description: This pie chart illustrates how casual riders are distributed across the different rideable types. The chart was created by filtering the dataset for Casual users and using rideable type to represent the distribution of rides. The pie chart shows that electric bikes account for the highest percentage of rides among casual riders, followed by classic bikes. This indicates that casual riders have a stronger preference for electric bikes compared with classic bikes.

Dashboard 1: Rideable Type Distribution: Members vs Casual Riders

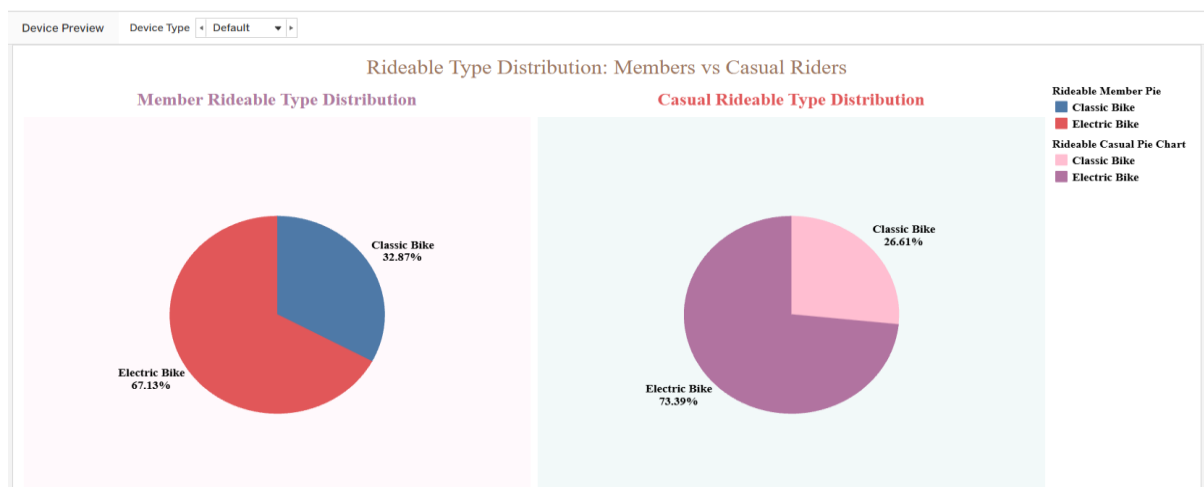


Fig. IX. C. 7. Rideable Type Distribution of Member and Casual Riders

Description: The first dashboard combines two visualizations: Member Rideable Type Distribution and Casual Rideable Type Distribution. These pie charts provide a comparative view of the distribution of different rideable types among member and casual riders, making it easier to identify differences in their rideable-type preferences.

Dashboard 2: Casual vs Member Comparative Analysis

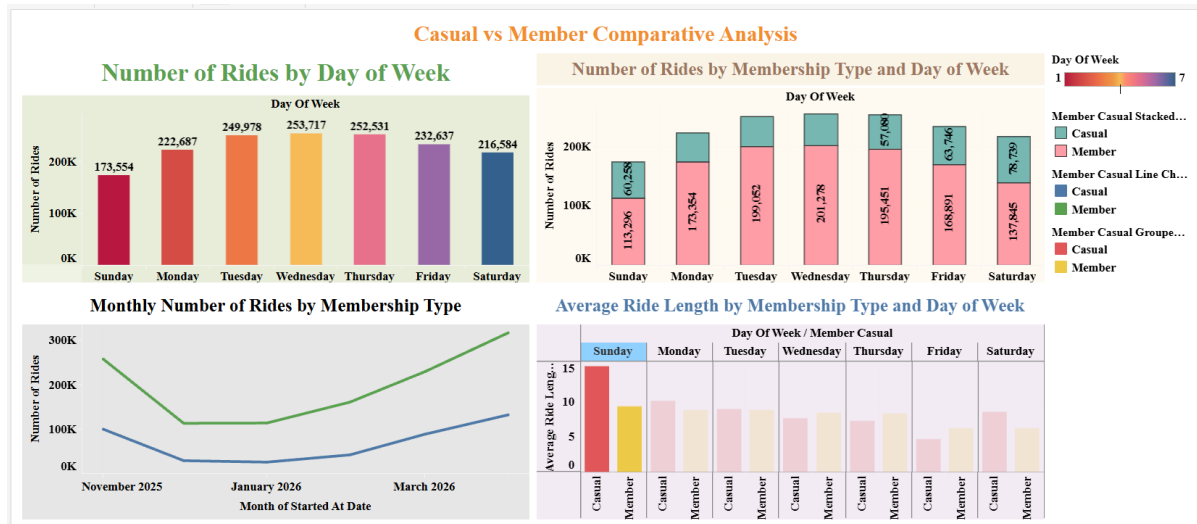


Fig. IX. C. 8. Comparative Analysis of Casual and Member Riders

Description: The second dashboard combines four visualizations: Bar Chart, Stacked Bar Chart, Grouped Bar Chart, and Line Chart. These visualizations provide a comparative analysis of casual and member riders and help identify differences in ride usage, rideable-type distribution, and riding patterns across the days of the week.

Tableau Story

A Tableau Story was created to present the analysis in a structured and interactive manner. The Story consists of **5 story points**, including the four individual visualizations and the dashboard. The main story points are:

- 1) Number of Rides by Day of Week
- 2) Number of Rides by Membership Type and Day of Week
- 3) Monthly Number of Rides by Membership Type
- 4) Average Ride Length by Membership Type and Day of Week
- 5) Rideable Type Distribution: Members vs Casual Riders

X. Phase 6: ACT

A. Final Conclusion

1. Members have significantly more rides than casual riders, accounting for nearly three times the number of casual rides during the analysis period.
2. Casual riders had a higher average ride length on Sunday, Monday, Tuesday, and Saturday, while members had a higher average ride length on Wednesday, Thursday, and Friday. Overall, Members recorded more rides during the middle of the week, while casual riders recorded more rides at the beginning and end of the week.
3. Both members and casual riders show a strong preference for electric bikes.

B. Recommendations

- Offer targeted membership promotions to casual riders, especially on weekends.
- Promote annual memberships by highlighting the benefits for frequent and longer rides.
- Focus marketing campaigns on electric bikes to attract casual riders and encourage membership conversion.